ZXP1009盘车控制箱
MODEL ZXP1009 CONTROL BOX FOR TURNING GEAR

使用说明书
OPERATING INSTRUCTIONS

编制：
校对：

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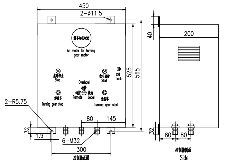
盘车控制箱用于控制盘车装置，使盘车装置手动启停。
This control box is used to control the turning gear device for its manual start-up and stop

1. 控制箱柜体
Body of the control box

柜体由冷镀钢板制成，内有接线端子排和各种逻辑器件。外型尺寸与安装尺寸见下面两张图，信号线和电缆线由柜体下方接入。防护等级IP52。箱体喷漆为RAL 7032，浅灰色。
The body is made of hard-wrought steel plate, inside which are fixed terminal blocks and various logical elements. Overall dimensions and mounting sizes are shown in the following figures. Both the holding wires and cables are led into the lower part of the box.



控制箱正面
Front

控制箱侧面
Side

2. 控制箱面板
Panel of the control box

盘车控制箱的操作由控制面板完成，控制面板上有一个转换开关、两个个操作按钮、二个指示灯、一个电流表。开盘车按钮：用于现场启动盘车装置；停盘车按钮：用于现场停止盘车；远控/检修/就地选择转换开关：用于选择自动操作还是手动操作，手动操作需要工作人员在现场或控制室启动或停止盘车，自动操作由控制系统自动启停盘车装置。二个指示灯为盘车电机启停指示。电流表用于指示盘车电机电流。

The control box is operated through the medium of the panel, on which are one switch ,three operating buttons, two indicator lamps and one ammeter. ON button is used to start the turning gear; and OFF button to stop the turning gear at the site. REMOTE/ LOCAL switch is used for a choice between automatic and manual operating modes. The turning gear will be started or stopped at the site or in control room by an operator in manual mode, or by the control system automatically in automatic mode. The two indicator lamps are for the indication of the turn gear start and turn gear stop. And, the ammeter is for the indication of the current of turning gear motor.

3. 接线端子及接线图
Connecting terminals and wiring diagram

进出控制箱的电缆都经接线端子连接，端子接线见端子接线图，下图是端子说明。
Cables in and out the control box are connected through the connecting terminals, as shown in the wiring diagram. The terminals are described as following.

控制回路端子排：
Terminal block for control circuit:

端子号（注1） Term.NO.		说明 Description	备注 Remarks
DZ1−1	DZ2−1	盘车电机启、停信号(闭合启动、断开停止) Start-up signal of HP oil pump	来自控制室信号 The signal from control room
DZ1−2	DZ2−2	盘车启动允许信号（组合信号，见注3） Allow starting the turining gear	
DZ1−3	DZ2−3	盘车电机启动信号给DCS Starting signal	去控制室信号 The signal to control room
DZ1−4	DZ2−4	电磁阀电源出线 The power supply for solenoid valve	去润滑电磁阀
DZ1−5	DZ2−5	电磁阀电源入线 The power supply for solenoid valve	供电方式详见注2
DZ1−6	DZ2−6	故障信号 The signal of fault	去控制室信号 The signal to control room
DZ1−7	DZ2−7	远程控制方式 Chose remotely control	
DZ1−8	DZ2−8	就地控制方式 Chose locally control	
DZ1−9	DZ2−9	检修控制方式 Chose Overhaul control	
DZ1−10	DZ2−10	盘车电机电流信号 4~20mA An meter for gear motor	

主回路端子排：
Terminal block for main circuit:

序号 S/N		说明 Description
1	R	电机电源进线 The inlet cable of power supply for the motor
2	S	
3	T	
4	U	电机电源出线 The outlet cable of power supply for the motor
5	V	
6	W	
7	N	零线

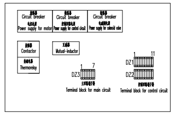
注1：端子排电源极性：DZ1为+或L、DZ2为N或−。

注2：当电磁阀电源与控制回路电源相同时无需外供电，从控制回路直接取电；当电磁阀电源与控制回路电源不相同时需外供电。

操作面板说明
Control switch introdufion for operationing panel

序号 NO.	说 明 Introduction	型号 MODEL	厂家 Manufacturer
SA1	就地/远控/停止选择开关（3位） Select local/remote/tomid(three bit)	见设备随机清单	施奈德
SB1	启动按钮 Local start	见设备随机清单	施奈德
SB2	停止按钮 Local shut-down	见设备随机清单	施奈德
HG1	盘车启动指示 Power indication	见设备随机清单	施奈德
HR1	盘车停止指示 Running indication	见设备随机清单	施奈德
KM1	接触器 Contactor	见设备随机清单	施奈德
QF1	断路器 Breaker	见设备随机清单	施奈德
QF2	自动开关 Switch	见设备随机清单	施奈德
QF3	自动开关 Switch	见设备随机清单	施奈德
KR1	热继电器 Thermorelay	见设备随机清单	施奈德
TAb1	电流互感器 Current transformer	见设备随机清单	正泰
EPA1	电流变送器 Current transmitter	见设备随机清单	厦门宇电

端子排、逻辑原理图见下图，所标的数字为端子排号：
Terminal block and logic pattern are shown as following, in which the digits are the number of terminal block:



4. 操作说明
Operating instructions

使用前仔细检查接线，现场来的信号及电源线按汽轮机厂要求和端子箱接线图连接好.。控制箱可以控制的电机功率见控制箱型号说明，通过热保护继电器的旋钮选择不同功率的电机，确信无误后，合上断路器开关。
Before the operation, carefully inspect the connecting cable.The cable for local signals and power supply are to be connected according to the steam turbine maker as well as to the wiring diagram of terminal box. The control box is capable to control over motors ranging refer model of control box, which can be chosen through the knob of a thermally protective relay.

手动盘车
Manual turning gear:

现场：将面板上投切开关（SA1SA1s）置于手动位置，汽机转子停下后手动将汽机盘车齿轮啮合上，按面板上启动按钮（SB1SB1），盘车电机启动。如果要停盘车，按面板上的停车按钮（SB2SB2）即可。
Local: Set the switch SA1 on the panel to LOCAL position. Once the turbine rotor stops rotation, force the tooth of turning gear to mesh in. And then, push the knob SB1 to start the turning gear motor. To stop the turning gear, push the stop knob SB2 on the panel.

控制室：将面板上投切开关（SA1SA1SA1）置于自动位置，汽机转子停下后在，在控制室将盘车控制信号投入，使盘车电机启动。将控制信号断开，盘车电机停止。在就地可以将面板上的开关（SB2）置于停止位置，使盘车电机停止。
Control room: Set the switch SA1 on the panel to AUTO position. Once the turbine rotor stops rotation, And then, in the control room, put into service the switch to start the turning gear motor, break the switch to stop the turning gear motor. Set the switch SB2 on the panel , to stop the turning gear motor.

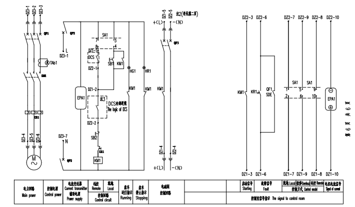
注意事项
Notice:

（1） 当汽机润滑油压低，顶轴油压没有建立和汽轮机转速高，禁止投入盘车装置，控制回路将禁止启动盘车电机。
It is forbidden to put the turning gear into service and start the turning gear motor through control circuit in the case that the turbine lubricating oil pressure is low, jacking oil pressure has not been built up and the turbine rotor speed is high.
（2） 当盘车电机电流过大，热保护继电器将保护动作，可按继电器上小按钮恢复。
In case of an excess current of turning gear motor, a thermal protection relay will function for protection, which can be reset by pushing a small button on the relay.
（3）如果没有顶轴油压信号，控制室信号D取消。
In case of no signal for jacking oil pressure, the signal C will cancel.

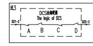
5. 控制箱型号
Model of control boxes:

ZXP1009-A-B

A:润滑电磁阀电源24VDC、110VDC、220VDC、220VAC
A:Power supply 24VDC 110VDC 220VDC 220VACfor solenoid valves for lubrication
B:盘车电机功率1.5KW、3.5KW、5.5KW、7.5KW、11KW、13KW等。
B:The power of motors for 1.5KW、3.5KW、5.5KW、7.5KW、11KW、13KW.
举例:ZXP1009-24VDC-7.5KW
电磁阀电压24VDC 盘车电机功率7.5KW
Exemple:ZXP1009-24VDC-7.5KW
Power supply 24VDC for solenoid valves for turning gear and lubrication
The power of motors for 7.5KW



注3



A: 润滑油压低停盘车 Turn gear stops at low lubricating oil press.
B: 盘车装置投入信号 Switch-in signal of turning gear
C: 汽轮机转速升至脱开转速 Turbine speeds up（如无脱开转速信号则取消）
D: 顶轴油压建立 Build-up of jacking oil press.(如无顶轴油压，则取消此信号)

转换开关SA1位置表示
Express of switch (SA1)

	60° 就地 Local	0° 检修 Overhaul	60° 远方 Remote
1−2	X		
3−4	X		
5−6		X	
7−8			X
9−10			X